

System Requirements Specification

Project: Fresh to Home eCommerce Website

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Version History

Version	Date	Description	Author
1.0	Feb 2026	Initial Draft	Business Analyst

1. Executive Summary

1.1 Purpose

The purpose of this document is to outline the functional and non-functional requirements for the "Fresh to Home" eCommerce website. This platform will enable the client, a supermarket based in Ernakulam, to sell groceries, vegetables, meat, and milk online.

1.2 Project Scope

Scope Area	Description	Scope Area
Platform	Responsive Website (Desktop & Mobile Web).	Platform
Inventory	Single Admin managing inventory for two specific physical branches.	Inventory
Key Differentiators	Same-day delivery logic (7km radius), Cryptocurrency payments (Binance), Subscription model.	Key Differentiators
Goal	Order daily essentials (groceries, vegetables, meat, milk) online.	Goal

2. System Overview

2.1 Proposed System

"Fresh to Home" will serve as a digital storefront for the physical supermarket. It is designed to capture customer data (Phone/Email) for direct marketing and streamline local delivery in Ernakulam. The system automatically assigns orders to the nearest of the two physical shops based on the customer's pin code.

2.2 Key Features

Feature	Description	Feature
Location-Based Access	Service is restricted to specific pin codes within a 7km radius of the shops.	Location-Based Access
Multi-Store Management	One Admin panel controls stock and orders for two physical branches.	Multi-Store Management
Smart Delivery	Orders are fulfilled by the nearest shop with same-day delivery options.	Smart Delivery
Subscriptions with Offers	Recurring order options for "all-month" items (e.g., daily milk) with special subscription discounts.	Subscriptions with Offers

Payment Flexibility	Integration of UPI, Cash on Delivery (COD), and Binance (Crypto) payments.	Payment Flexibility
Scheduled Delivery	Customers can choose specific delivery time slots with cut-off logic for same-day delivery.	Scheduled Delivery

2.3 Target Audience

Dimension	Details	Dimension
Primary Users	Families and households residing within a 7km radius of the two physical store locations.	Primary Users
Demographics	Ages 25–60; individuals managing household groceries and daily essentials.	Demographics
Tech Proficiency	Low to Moderate. UI must be simple, localized, and intuitive.	Tech Proficiency
Needs	High reliability for daily perishables (Milk/Curd) and home delivery convenience.	Needs
Payment Preference	Mixed: UPI, Binance (Younger), COD (Older).	Payment Preference

3. Functional Requirements (Page-by-Page UI Specs)

3.1 Global UI Elements

Component	UI Elements & Requirements
Header Navigation	• Logo: Clickable, redirects to Homepage. • Location Indicator: Displays selected Pin code (e.g., "Delivering to 682001"). • Search Bar: Placeholder "Search for milk, veg, meat...". • Account Icon: Dropdown (Login/Profile/Orders/Subscriptions). • Cart Icon: Floating icon with item count badge.
Footer	• Quick Links: Home, About Us, Contact Us. • Categories: Vegetables, Meat, Dairy, Grocery. • Legal: Privacy Policy, Terms of Service, Refund Policy.

	• Social Media: Icons for Instagram / Facebook / WhatsApp. • Copyright: "© 2026 Fresh to Home Supermarket".
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3.2 Landing & Location Check

Goal: Strictly enforce the 7km delivery radius before allowing browsing.

Component	Specifications & Logic
UI Components	• Modal Popup: Non-dismissible on first visit. • Input Field: "Enter your Pin code". • Button: "Check Availability". • Hero Image: High-quality image of fresh vegetables.
Logic	1. User enters Pin code. 2. System calculates distance to Store A and Store B. 3. Scenario A (Within 7km): Success message -> Redirect to Homepage. Store Cookie. 4. Scenario B (Outside 7km): Error message "Sorry, we do not deliver to this area yet." -> Prevent access.

3.3 Registration & Login Page

Goal: Capture user data for marketing.

Component	Specifications & Logic
UI Components	• Tabs: Login / Sign Up. • Input Fields (Sign Up): Name, Mobile Number (Mandatory), Email ID (Mandatory). • Input Fields (Login): Mobile Number or Email. • Button: "Send OTP". • OTP Input: 4-digit or 6-digit field.

Logic	• Validates mobile number format (10 digits). • Verifies Email format. • On successful OTP verification, creates user record in DB and logs in.
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3.4 Homepage

Goal: Showcase current offers and top categories.

Component	Specifications & Logic
UI Components	• Hero Slider: Carousel of 3-4 banners (e.g., "50% off on Chicken"). • Shop Offers Section: Horizontal scroll of items with specific discounts. • Category Grid: Icons for Veg, Fruits, Meat/Fish, Dairy, Grocery, Bakery. • "Deal of the Day": Timer-based section showing 3 products with discounts. • Floating WhatsApp Button: Bottom right corner.

3.5 Product Listing Page (PLP)

Goal: Easy browsing and filtering.

Component	Specifications & Logic
UI Components	• Filters: Category (Veg/Non-Veg), Price, Dietary. • Product Card: Image, Title, Price Display (Strikethrough MRP & Offer Price), Unit Selector. • Action Button: "Add" (changes to +/-) OR "Subscribe".
Logic	• Only displays items in stock at the <i>assigned</i> branch. • "Out of Stock" items are greyed out with "Notify Me".

3.6 Product Detail Page (PDP)

Goal: Detailed info and subscription toggle.

Component	Specifications & Logic
UI Components	• Image Gallery: High-res zoomable images. • Info Section: Title, Description, Nutritional Info. • Subscription Toggle: - Option A: "One-time Purchase" - Option B: "Subscribe & Save" (% Off) • Frequency Selector: Daily, Alternate Days, Weekly (only if subscribe selected). • Sticky Footer (Mobile): "Add to Cart" button fixed at bottom.

3.7 Cart Drawer / Page

Goal: Review and upsell.

Component	Specifications & Logic
UI Components	• Item List: Image, Name, Quantity Counter, Total Price. • Coupons Section: "Apply Coupon" input field. • Bill Details: Item Total, Delivery Fee, Taxes, Total Payable. • Checkout Button: "Proceed to Checkout".
Logic	• Checks stock one last time before checkout. • Validates Minimum Order Value (if any).

3.8 Checkout Page

Goal: Secure Payment and Scheduling.

Step	UI Components & Logic
Step 1: Address	• List of saved addresses. • "Add New Address" with Google Maps pin integration.
Step 2: Delivery	• Date Picker: Today, Tomorrow, Day After. • Time Slots: 8-10 AM, 10-12 PM, 4-6 PM, 6-8 PM. • Logic: Disable slots past current time + 2hr buffer. Same-day cut-off at 5 PM.
Step 3: Payment	• UPI: (GPay, Phonepe logos). • Binance Pay: (Crypto Icon). • Credit/Debit Card. • Cash on Delivery: (Disabled if order > ₹2000).
Final Action	• "Place Order" button.

3.9 My Account & Subscription Dashboard

Goal: Self-service management.

component	Specifications & Logic
Overview	Name, Phone, Email, Edit Profile.
My Orders	List of past orders with "Reorder" button.
Subscriptions	• List of active subscriptions. • Calendar View: Visual calendar of delivery dates. • Actions: Pause / Cancel Subscription. • Wallet: Balance check and "Add Money".

4. Admin & Store Management Functionalities

4.1 Central Admin Dashboard & Reports

Feature	Requirements
Key Metrics	Display Total Sales (Today/Weekly), Total Orders, Active Subscriptions, Low Stock Alerts.
Data Export	- Fields: Customer Name, Phone, Email, Last Order Date. - Function: "Export to CSV" button for marketing.
Sales Reports	Breakdown sales by Payment Method (UPI/Binance/COD) and Branch (Store A/Store B).

4.2 Branch & Inventory Management

Feature	Requirements
Store Selector	Toggle between "Branch A" and "Branch B" views.
Product Mgmt	- Fields: Name, Category, MRP, Selling Price, SKU, Unit, Image. - Stock Input: "Current Stock Level" specific to selected branch.
Bulk Update	CSV upload option to update stock levels for 500+ items daily.
Alerts	Set "Low Stock" limits (e.g., alert when Milk < 10 pkts).

4.3 Order Management & Routing

Feature	Requirements
Order List	Columns: Order ID, Customer Name, Pin code, Total Amount, Status, Delivery Slot.
Status Management	Dropdown: <i>Pending -> Packed -> Out for Delivery -> Delivered -> Cancelled.</i>
Routing	- System defaults to "Nearest Store". - Manual Override: Admin can "Transfer Order" to the other branch if out of stock.

4.4 Promotions & Offers Management

Feature	Requirements
Coupons	Code (e.g., WELCOME50), Type (Flat/%), Min Cart Value, Expiry Date.
Product Offers	Select Product -> Set "Offer Price" (System calculates % off MRP).
Banners	Upload interface for Homepage Banners with scheduling (Start/End Date).

4.5 Subscription Management

Feature	Requirements
Dispatch List	One-click generation of "Morning Delivery Sheet" for subscription items.
Subscriber List	View active subscribers with pause/resume status.

5. Detailed User Stories

5.1 Customer User Stories

ID	User Story Name	As a...	I want to...	So that...	Pre-Conditions	Acceptance Criteria
US-01	Location Validation	New Visitor	Enter my Pin code upon landing	I know if delivery is available.	User visits homepage for first time.	1. Popup prompt appears. 2. Validates against 7km radius. 3. Blocks checkout if outside zone.
US-02	Registration	Customer	Register using Phone & Email	I can create an account and get updates.	User is on Registration page.	1. Phone & Email mandatory. 2. OTP verification sent.

						3. Registration required for order.
US-03	Subscription	Regular Shopper	Subscribe to Milk with offer	I save money and automate orders.	User viewing eligible product.	1. "Subscribe" button visible. 2. Frequency selection (Daily/Weekly). 3. Discount (e.g., 5%) applied. 4. Auto-billing setup.
US-04	Product Search	Customer	Search for "Tomato" or "Chicken"	I find items quickly.	User on any page.	1. Search bar in header. 2. Results filtered by nearest store. 3. Category filters available.
US-05	Scheduling	Customer	Select Delivery Time Slot	I am home to receive fresh produce.	User at checkout.	1. Date/Time picker displayed. 2. Same-day cut-off enforced. 3. Available slots listed.
US-06	Crypto Pay	Crypto User	Pay using Binance Pay	I use my crypto for daily needs.	User at payment step.	1. "Binance Pay" option listed. 2. Secure redirect to gateway.

						3. Auto-confirmation received.
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5.2 Admin & Store User Stories

ID	User Story Name	As a...	I want to...	So that...	Pre-Conditions	Acceptance Criteria
US-07	Branch management	Super Admin	Manage inventory for 2 branches	Stock is accurate per location.	Admin logged in.	1. Branch toggle available. 2. Independent stock updates.
US-08	Auto Routing	System	Route orders to nearest store	Delivery is efficient.	Order placed.	1. Distance calculated. 2. Order assigned to nearest store. 3. Manual override available.
US-09	Data Export	Marketing Mgr	Export customer lists	I can run SMS/Email campaigns.	Admin logged in.	1. "Export" button available. 2. Standard CSV format. 3. Filter for "Active Users".
US-10	Promotions	Admin	Create Festival Discounts	I boost sales during holidays.	Admin logged in.	1. Rules based on % off MRP.

						2. Schedule start/end dates. 3. Auto-display on Homepage.
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6. Technology Stack

Component	Technology	Description
Frontend	React.js	For building a responsive, dynamic user interface (SPA).
Styling	Tailwind CSS	Utility-first CSS framework for rapid UI development.
Backend	Node.js with Express.js	Scalable server-side architecture.
Database	MongoDB	NoSQL database for flexible product schema and user data.
Cloud Hosting	AWS / Digital Ocean	For hosting application and database with auto-scaling.
Version Control	Git / GitHub	For source code management.

7. System Integrations

Service Type	Integration	Purpose & Flow
Payment Gateway	UPI / Card Gateway	• Purpose: Process standard currency transactions. • Method: Standard API integration.
Payment Gateway	Binance Pay	• Purpose: Accept Cryptocurrency payments. • Flow: User selects "Binance" -> QR Code -> Webhook confirmation.
Payment Gateway	Cash on Delivery	• Purpose: Allow cash payments.

		• Logic: Enabled based on Order Value caps (Max ₹2000) or Pin code.
Communication	SMS Gateway (Twilio / MSG91)	• Triggers: OTP for Login, Order Confirmation, "Out for Delivery" alerts.
Communication	Email Service (SendGrid)	• Triggers: Order Receipt, Welcome Email, Subscription Renewals.
Location	Maps API (Google Maps / Map box)	• Purpose: Validate 7km delivery radius. • Logic: Calculates distance between user pin code and store.

8. Non-Functional Requirements

Requirement Category	Description
Platform	Website only (Responsive HTML/React/Node stack).
Performance	Website must load quickly (<3 seconds), especially for image-heavy listings.
Security	• Secure storage of customer phone/email data. • Secure processing of UPI and Binance transactions.
Scalability	System should handle high traffic (500+ concurrent users) during festival seasons.
Uptime	99.9% availability during peak ordering hours (6:00 AM – 10:00 AM).

9. Assumptions & Constraints

Category	Details
Service Area	Delivery is strictly limited to Ernakulam within the defined 7km radius.
Logistics	The client manages their own delivery fleet; the system only schedules the time slots.
Payment	User must have a valid Binance account for crypto payments.

10. Conclusion

The "Fresh to Home" eCommerce platform represents a strategic digital transformation for the client, designed to bridge the gap between physical grocery shopping and digital convenience in Ernakulam. By integrating hyper-local delivery logic with flexible payment solutions like Binance and UPI, the system addresses the immediate business need for customer data acquisition while providing a seamless ordering experience for users.

This System Requirements Specification outlines a robust foundation for development, focusing on critical features such as multi-branch inventory management, automated order routing, and a recurring subscription model. The proposed system is scalable and tailored to the local market, ensuring operational efficiency and long-term business growth. We are confident that this specification will guide the successful deployment of a high-quality, user-friendly platform that meets all stakeholder expectations.

11. Disclaimer

This System Requirements Specification (SRS) document contains confidential and privileged information intended solely for the stakeholders of "Fresh to Home". Any unauthorized review, use, disclosure, or distribution of this document is strictly prohibited.

The actual project scope, deliverables, features, and timeline may be subject to change based on evolving business requirements, technical feasibility, or regulatory updates. Specifically, features involving cryptocurrency payments (Binance) are subject to local financial regulations and compliance requirements, which should be verified independently by the business owners. The development team assumes no liability for external factors affecting third-party integrations or legal compliance.